

RESEARCH ARTICLE

Spillover Effects From Health Information Systems Integration: Evidence From Prescription Drug Monitoring Programs

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Received: 23 October 2023 | **Revised:** 12 December 2024 | **Accepted:** 22 February 2025

Handling Editors: Meng Li, Paul A. Pavlou, Elliot Bendoly, Jeffery Smith

Funding: Yufei Shen acknowledges funding support from FCT (UIDB/00124/2020, UIDP/00124/2020, UID/00124, POR Lisboa and POR Norte (PINFRA/22209/2016)). Dorothy Jiang acknowledges funding support from the Behavioral Research Assistance Grant, the C. T. Bauer College of Business, University of Houston. Dimitrios Andritsos acknowledges funding support from the HEC Paris Foundation, Qatar Foundation, and Labex Ecodec: Investissements d'Avenir (ANR-11-IDEX-0003/Labex Ecodec/ANR-11-LABX-0047). Xitong Li acknowledges funding support from the HEC Paris Foundation and research fellowship from Hi! PARIS, a joint research center between HEC Paris and Institut Polytechnique de Paris.

Keywords: difference-in-differences | health information systems | healthcare operations | prescription drug monitoring programs | spillover effect | supplied opioid quantity

ABSTRACT

To ease prescribers' burden in using statewide Prescription Drug Monitoring Programs (PDMPs), many US states have moved to better integrate PDMP data with patients' electronic health records (EHRs). While such PDMP-EHR integration is likely to reduce local formal prescription opioid consumption, in this paper, we explore the possibility that it induces individuals to seek opioids across state borders and/or at the local black market. We find that for a focal county, its neighboring state's PDMP-EHR integration leads to an increase in the focal county's consumption of prescription opioids, an undesirable cross-border spillover effect. Moreover, we find that this spillover effect is mitigated when prescribers in the focal county have access to the neighboring state's PDMP data via interstate PDMP data sharing. We also find nuanced evidence that a state's PDMP-EHR integration might induce an increase in its black-market prices of prescription opioids. While vertical information integration in a region (i.e., a state's PDMP-EHR integration) may fulfill its intended benefit, our findings suggest it may also result in a spillover effect on neighboring areas. Notably, a complementary horizontal information integration (in this case, interstate PDMP data sharing) could mitigate such an undesirable spillover effect. We discuss implications for the design of care delivery processes, particularly prescription processes, that rely on the studied health information systems.

1 | Introduction

Prescription opioids, such as OxyContin, are widely used for pain management. However, their misuse can result in serious health problems (e.g., addiction and mortality), impose

significant burdens on health systems, and generate substantial social and legal repercussions (Government Accountability Office 2018). In recent years, opioid availability and opioid-related deaths (ORDs) have risen across Organization for Economic Co-operation and Development (OECD) countries

Summary

- State-level policies that enhance the monitoring of patients' consumption of prescription drugs, such as the integration of Prescription Drug Monitoring Programs (PDMPs) into electronic health records, may lead to unintended cross-state spillover effects, increasing opioid prescription demand in neighboring states.
- Interstate PDMP data sharing can help mitigate these spillover effects.
- Such integration could also result in modest increases in opioid prices within local black markets.

(OECD 2019). Among the OECD countries with available data, the surge in ORDs has been “most pronounced in the United States, Canada, Sweden, Norway, Ireland, and England and Wales”. In the United States (US), which is the focus of this study, opioid misuse has reached such a critical level that the US government officially acknowledged the presence of an opioid overdose crisis and declared it a “national health emergency” on October 26, 2017 (Government Accountability Office 2018). It continues to represent a significant public health challenge.

Enhanced surveillance of prescribing practices and patient behavior is essential to address this challenge (OECD 2019). Consequently, several North American countries, Australia, and parts of Europe have introduced electronic Prescription Drug Monitoring Programs (PDMPs) (Islam and McRae 2014), health information systems that track the prescription and dispensation of controlled substances. For example, in the United States, PDMPs operate at the state level, recording transaction data when patients receive prescription opioids from pharmacies. Authorized prescribers access this data through PDMP web portals to review patients' consumption histories, drug types, quantities, and prescribed durations.

This study examines the integration of PDMP data into the Electronic Health Record (EHR) (hereafter simply referred to as PDMP-EHR integration), a form of *vertical* information integration promoted by several US states. Formally, vertical information integration refers to a top-down or bottom-up flow of information between organizations (or different levels of an organization) that is enabled via appropriate information systems integration (Hendricks et al. 2007). PDMP-EHR integration improves prescribers' access to PDMP data directly within their EHR interface, incorporating features like single sign-on, automated PDMP queries, and summarized patient records.¹ Consequently, it reduces the time and effort to retrieve information, supporting more efficient clinical decisions. This addresses a major barrier, as prescribers often feel PDMP access is cumbersome and misaligned with clinical workflows. A 2014 national survey found that only 53% of primary care physicians used PDMPs due to such burdens (Rutkow, Turner, et al. 2015). The pilot study conducted by the US Centers for Disease Control and Prevention (CDC) in Illinois showed a 145-fold increase in PDMP data requests per prescriber following integration (US CDC 2017).

Recent evidence suggests that PDMP-EHR integration can increase prescriber queries of PDMP data (Neprash et al. 2022) and may decrease opioid-related morbidity and mortality (Wang 2021). Our study aims to complement existing research by examining *indirect*, possibly unintended effects of PDMP-EHR integration. Specifically, we investigate whether PDMP-EHR integration may drive state populations to seek opioids across state borders or within their state's black market. These effects, largely neglected in the literature on PDMP-EHR integration, are critical for evaluating the effectiveness and externalities of PDMP implementations.

The relationship between a focal county's prescription opioid consumption and a neighboring state's PDMP-EHR integration is complex. On the one hand, PDMP-EHR integration in a neighboring state might encourage opioid consumers to strategically “doctor shop” for prescriptions across state borders in the focal county,² where providers might be unaware of consumers' prescription history (Cepeda et al. 2013). Such movement could increase the focal county's prescription opioid consumption. However, a heightened awareness of the opioid epidemic might lead prescribers in the focal county to adopt more conservative prescribing practices when treating out-of-state patients to mitigate legal or medical risks. This cautious approach could reduce opioid prescriptions, introducing ambiguity in the overall impact of a neighboring state's PDMP-EHR integration on opioid consumption in the focal county.

On the other hand, a neighboring state's PDMP-EHR integration might drive consumers to their local illicit markets, potentially leaving focal county consumption patterns unchanged while raising local black market prices for opioids due to supply-demand dynamics (KC et al. 2022; Mallatt 2022). The risks associated with black market purchases suggest this would occur only if the benefits outweigh the costs. Significant price disparities between legal and street prices may also deter illicit purchases. For instance, the cost of OxyContin (15 mg), a popular prescription opioid, is \$6.52 per pill (price without insurance), whereas the street price for the same drug is \$15 in 2024.³ Additionally, federal and state laws classify unauthorized possession or sale of controlled substances as felonies (Schnell 2022), which may further discourage black market transactions. Therefore, the overall effect of PDMP-EHR integration on the black market for opioids remains ambiguous.

Understanding such indirect effects of PDMP-EHR integration calls for empirical examination of the following two research questions:

1. Cross-border spillover effect: Does a neighboring state's PDMP-EHR integration lead to any spillover effect on the consumption of prescribed opioids in a focal county?
2. Black-market spillover effect: Does a focal state's PDMP-EHR integration lead to any spillover effect on its local “black market” for prescription opioids?

For the first question, we exploit the fact that different states have implemented PDMP-EHR integrations at different time points and employ a staggered difference-in-differences (DID) framework to estimate the cross-border spillover effect. Our

empirical results show that prescribed opioids' supply in a focal county (our proxy for consumption), on average, increases significantly (by approximately 2%) when at least one of its neighboring states has implemented PDMP-EHR integration, suggesting an unintended, undesirable cross-border spillover effect (Meinhofer 2018). Moreover, we find that the cross-border spillover effect is stronger when the focal county is geographically closer to the border with the neighboring state implementing PDMP-EHR integration and the neighboring state has a higher-than-average EHR adoption rate at its hospitals. These findings provide support to the argument that the cross-border spillover effect is likely driven by patients' cross-border "doctor shopping" behavior. For the second question, we examine the impact of PDMP-EHR integration on local black-market prices for prescription opioids and find that in states with higher-than-average EHR adoption rates, PDMP-EHR integration is associated with a 1.6% increase in street prices.

Motivated by these findings, we investigate if *horizontal*, cross-state information integration—where prescribers in the focal county access the neighboring state's PDMP data—can moderate the identified spillover effects. Unlike vertical information integration, horizontal information integration occurs at similar organizational levels, enabling peer collaboration, knowledge sharing, and coordination, which may support more informed decision-making (Vickery et al. 2003). In our setting, horizontal information integration could aid prescribers in making more informed decisions when prescribing for visiting patients. Recognizing the importance of this issue, many states have established legal agreements to share PDMP data through a national data-sharing network. However, state participation in such a network could also increase price pressure in local illicit opioid markets by limiting cross-border doctor shopping opportunities. Hence, we also examine the subsequent follow-up questions:

3. Cross-border spillover effect mitigation: How is the cross-border spillover effect moderated by the focal county's prescribers' access to the neighboring state's PDMP data via interstate PDMP data sharing?
4. Black-market spillover effect amplification: How is the black-market spillover effect moderated by the focal state's participation in interstate PDMP data sharing?

We find that the undesirable cross-border spillover effect is mitigated (by approximately 40%) when prescribers in the focal county have access to the neighboring state's PDMP data via interstate PDMP data sharing, which suggests a complementarity between within-state *vertical* PDMP data integration and across-state *horizontal* PDMP data integration. We do not find evidence that state participation in interstate PDMP data sharing moderates the local black-market spillover effect.

Our paper, at the interface of Operations Management and Information Systems, contributes to the literature on information technology management (Heim et al. 2021), and specifically to the sub-stream that focuses on health information technologies (Gardner et al. 2015). In particular, we identify implications arising from integrating different Health

Information Systems (HISs)—namely, PDMPs and EHRs—that occur in the context of health policy implementation. We demonstrate that vertical integration of these systems can lead to unintended cross-border spillover effects, driven by regional variations in implementation. To our knowledge, this is one of the first studies to identify such regional spillover effects resulting from PDMP-EHR integration, aligning with recent research that reports unintended local consequences of mandated PDMP use, such as shifts toward heroin use (KC et al. 2022).

Our findings highlight the operational complexity of controlling prescription opioid flows and related behaviors, suggesting that regional vertical HIS integration discrepancies can undermine health policy objectives and decision-making effectiveness. However, we demonstrate that horizontal information integration could mitigate such regional mismatches in policy implementation. Our results underscore the complementarity between vertical and horizontal HIS integration, providing insights for effectively operationalizing health policies. In our conclusion, we offer recommendations for the design of care delivery and prescription processes that rely on HIS integration.

2 | Literature Review and Hypothesis Development

In the realm of healthcare delivery, information systems play a pivotal role in enhancing the integration of patient data, thereby aiding clinical decision-making and patient outcomes (Dobrzykowski and Tarafdar 2015). Integrating PDMP data into EHRs exemplifies this by facilitating access to comprehensive medication histories. EHRs serve as the technological foundation for sharing patient health information (e.g., prior procedures, tests, and medical history) within and across organizational boundaries (Wani and Malhotra 2018). Below, we begin by drawing on the vast literature streams in operations management and information systems that have extensively studied and developed the theoretical foundations of information integration to then develop our hypotheses around the likely spillover effects of vertical and horizontal integration of PDMP data.

2.1 | Vertical and Horizontal Information Integration

In addition to healthcare, several industries rely on information technology for vertical information sharing. For instance, in manufacturing, Enterprise Resource Planning (ERP) systems facilitate the flow of production data from shop floor operators to executive management, enhancing operational efficiency and strategic planning (Hendricks et al. 2007). In education, Learning Management Systems (LMS) transmit student performance metrics to administrators, supporting curriculum development and policy formulation (Alavi and Leidner 2001). In supply chains, cloud-based platforms allow organizations to share real-time data—such as demand forecasts, inventory levels, and shipment status—enhancing coordination, efficiency, and resilience (Harapko 2023).

A recurring theme in this literature is that vertical information integration improves communication and decision-making across hierarchical levels, promoting transparency (Ren et al. 2023), accountability (Cai et al. 2010), and responsiveness (Devaraj et al. 2007). Nevertheless, vertical integration may also lead to unintended negative effects. In supply chains, for example, integration can create inflexibility by fostering interdependencies, potentially slowing an organization's response to change (Das et al. 2006) or it may introduce risks of opportunistic behavior, such as a supplier charging higher prices after detecting increased hospital demand through inventory information sharing (Chen et al. 2013).

In contrast to vertical information integration, which facilitates the flow of information between hierarchical levels, horizontal information integration connects similar organizational levels through peer-to-peer system integration. As such, it enhances knowledge sharing, coordination, and more informed decision-making (Vickery et al. 2003). In the context of policy design and execution, different forms of horizontal information integration have played a critical role in helping US states coordinate their efforts when dealing with issues of fraud, consumer protection, or the flow of controlled goods. Such issues might arise from regional policy differences that individuals might attempt to exploit. For instance, the National Motor Vehicle Title Information System is a database that allows states to share vehicle title and history data to prevent title fraud and maintain accurate records.⁴ Closer to our setting, the National Instant Criminal Background Check System enables states to share information on individuals prohibited from purchasing firearms, preventing them from bypassing local gun laws by buying firearms in states with more lenient regulations.⁵

In our setting, PDMPs are state-run electronic databases that collect and monitor information on the prescribing and dispensing of controlled substances to individuals. Research on the effectiveness of initial PDMP implementations yielded mixed results, with some studies reporting reductions in opioid prescriptions and others finding minimal impact, possibly due to differences across states in program design, utilization, and enforcement (Buchmueller and Carey 2018). To improve effectiveness, many states now require prescribers to query PDMPs, ensuring access to comprehensive patient data before prescribing opioids. These mandates have generally led to reduced opioid prescribing and better clinical decision-making, showing a more consistent impact compared to voluntary PDMP use (e.g., Buchmueller and Carey 2018; Meinhofer 2018).

Integration of PDMP data in the EHR is a more recent enhancement. It is a form of vertical information integration aiming to further improve clinical decision-making by offering prescribers a comprehensive view of patients' prescription histories directly within their workflows. Similarly, horizontal information integration through interstate PDMP data sharing emerged from recognizing the limitations of initial PDMPs, which focused solely on in-state prescription records. Lacking access to out-of-state prescribers' data could potentially hinder the identification of patients' doctor shopping across state lines, offering opportunities for drug-seeking individuals to

exploit. This limitation was addressed in August 2011 with the launch of PMP InterConnect—a national PDMP network enabling data sharing across states, initially involving Indiana, Ohio, and Virginia. By April 2022, 52 PDMPs were sharing data through PMP InterConnect, facilitating nearly 1 billion monthly transactions.⁶

Table A1 offers a comprehensive overview of prior research examining the impact of various PDMP implementations and functionality enhancements. It also highlights the unanswered questions in the literature, which form the foundation of our paper's contributions to the large body of research on PDMP.

2.2 | Cross-Border Spillover Effect

Prior research on vertical and horizontal PDMP data integration has primarily focused on direct within-state impacts, with limited evidence on broader consequences. Regarding prescriber behavior, the randomized clinical trial study of Neprash et al. (2022) reports that PDMP-EHR integration increases prescribers' monthly PDMP queries by 60%, whereas Hartung et al. (2024) find that it correlates with a 2.7-fold rise in monthly PDMP queries. Regarding patient health outcomes, Wang (2021) analyzes nationwide state-level data and finds that PDMP-EHR integration reduces opioid-related morbidity but does not significantly affect mortality. Research on the impact of horizontal PDMP data integration shows limited within-state effects, with null results on opioid prescribing for noncancer chronic pain (Lin et al. 2019) and opioid-related mortality and morbidity (Wang 2021).

However, experience from locally differentiated policy implementations, such as in alcohol regulation, cannabis control, tobacco sales, and gun control, suggests that effective policy evaluation should consider both local impacts and potential unintended spatial spillovers. Stricter local regulations often drive cross-border behaviors that undermine policy goals. For example, consumers may travel to neighboring regions with more lenient alcohol and tobacco laws, reducing local revenue and complicating public health efforts (Goli et al. 2024; Harding et al. 2012). Similarly, cannabis legalization in one state can increase cannabis-related activities in neighboring states, creating enforcement challenges (Hao and Cowan 2020). In gun control, stricter laws lead to more firearm purchases in states with lenient regulations, exacerbating gun trafficking (Knight 2013).

This discussion of controlled goods' settings suggests that vertical PDMP-EHR integration in a neighboring state may make it harder for individuals seeking opioids to obtain prescriptions there, potentially driving them to cross state lines (e.g., to the focal county) where their prescription histories are inaccessible to prescribers. This migration can lead to higher opioid consumption in the focal county, where these prescriptions are likely filled (Thornton 2017). Interstate data sharing through horizontal PDMP integration via PMP InterConnect can reduce information asymmetry between patients from neighboring states and prescribers in the focal county, potentially mitigating spillover effects from neighboring states' PDMP-EHR integration. Hence, this discussion suggests the following two hypotheses:

Hypothesis 1. (Cross-border spillover effect of vertical PDMP information integration): The enactment of vertical PDMP information integration in a neighboring state is associated with an increase in the supplied opioid quantity of a focal (border or interior) county.

Hypothesis 2. (Mitigating role of horizontal PDMP information integration): The increase in the supplied opioid quantity of a focal (border or interior) county associated with the enactment of vertical PDMP information integration in a neighboring state is reduced if prescribers in the focal county have access to the neighboring state's PDMP data.

2.3 | Black-Market Spillover Effect

Traveling across state borders to procure opioids under higher presumed anonymity may not be a desirable option, especially if it entails traveling long distances (Cepeda et al. 2013). This suggests that reduced availability of prescription opioids due to PDMP-EHR integration could push patients to consider non-medical sources, such as local black markets. Recent studies indicate that PDMP mandates may lead to undesirable substitution effects; for instance, KC et al. (2022) found that such mandates may increase heroin-related deaths. Thus, PDMP-EHR integration might also increase pressure on black-market prices for prescription opioids.

We focus our analysis on black-market prices for prescription opioids as, among available data, they are the most reliable indicator and are consistent with prices provided by law enforcement or collected from darknet online marketplaces (Dasgupta et al. 2013). Purchased quantities—our ideal measure—are not available in a reliable way due to the illicit nature of the transaction. Drug availability in the black market heavily depends on illicit trade activities and local law enforcement regulations (Grzybowski 2004).

The integration of PDMP data into EHRs enhances healthcare providers' ability to monitor and curb opioid misuse, likely leading to stricter prescription practices (Gardner et al. 2015). This increased scrutiny can reduce opioid availability through legal channels, which may raise demand in the black market (KC et al. 2022). As the supply of legally obtained opioids diminishes, black market sellers can exploit the scarcity by raising prices, driven by (i) higher demand, which is relatively price inelastic (Grzybowski 2004) and (ii) reduced competition from legitimate sources. Consequently, vertical PDMP-EHR integration could unintentionally elevate black market opioid prices, reflecting basic supply and demand dynamics in illicit markets. Horizontal PDMP data integration, by facilitating cross-state identification of opioid prescription abuse, further limits opportunities for individuals to exploit system gaps. This additional constraint on legal access could amplify the upward pressure on black market prices due to increased scarcity, suggesting that horizontal integration may reinforce the impact of vertical integration on black market prices.

This discussion leads us to the following two hypotheses:

Hypothesis 3. (Black-market spillover effect of vertical PDMP information integration): A focal state's enactment of

vertical PDMP information integration is associated with an increase in the price of prescription opioids sold in the focal state's illicit markets.

Hypothesis 4. (Amplifying role of horizontal PDMP information integration): The increase in the price of prescription opioids sold in the focal state's illicit markets associated with the enactment of vertical PDMP information integration in the focal state is amplified if the state participates in cross-state PDMP data sharing.

3 | Data and Variable Definitions

We constructed two unique datasets for our empirical analysis. First, to examine the cross-border spillover effect, we built a panel dataset covering 2922 US counties from 2008 to 2017, with Nebraska being the first state to implement PDMP-EHR integration in 2011. Following prior literature (Wang 2021), we chose 2017 as the end year because, on October 26, 2017, the US administration declared the opioid crisis a “national health emergency,” leading to intensified policy interventions that could complicate our analysis (Government Accountability Office 2018). Our dataset includes counties from all US states except Alaska and Hawaii, which are not contiguous with other states. Second, to examine the black-market spillover effect, we collected a transaction-level nationwide dataset of illicit opioid purchases spanning from 2013 (the dataset's first year) to 2017.

3.1 | Dependent Variables: Supplied Opioid Quantity (SOQ) and Black Market Prices

Measuring legal purchases of prescription opioids: To test our first two hypotheses and examine the association between a neighboring state's PDMP-EHR integration and consumption for prescription opioids in the focal county we use SOQ_{it} (supplied opioid quantity) as our dependent variable. It is measured as the total grams of morphine milligram equivalent (MME) of controlled opioid substances supplied in county i in year t . We collect this from the Drug Enforcement Administration's Automation of Reports and Consolidated Orders System (ARCOS). ARCOS is a drug reporting system that monitors the sales of controlled substances through commercial drug supply chain channels to prescribers or points of sale (e.g., practitioners, pharmacies, hospitals, clinics etc.) but does not contain sales to ultimate users (i.e., patients). Thus, we use SOQ_{it} , which captures the legal supply of prescription opioids, as a proxy for end users' aggregate demand.

Following the literature on PDMP (Meinhofer 2018), we analyze eight commonly abused opioid active ingredients: codeine, fentanyl, hydromorphone, meperidine, morphine, oxycodone, methadone, and oxymorphone. To make the different opioid drugs comparable, we follow CDC guidance and convert them into a unified MME-based quantity. The total MME sum across these ingredients serves as our dependent variable, following prior research (Meinhofer 2018). We convert SOQ_{it} from the 3-digit zip code level to the county level using 2010 Census population proportions.

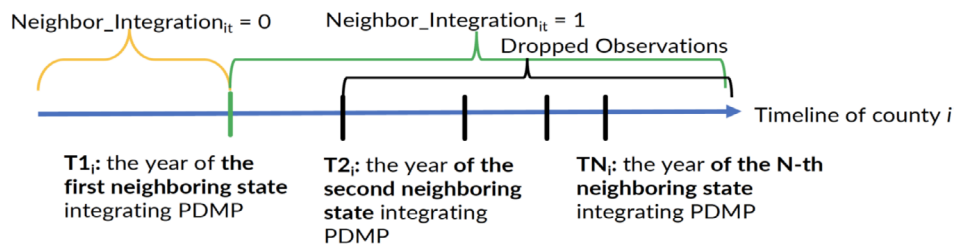


FIGURE 1 | Illustration of treatment definition and sample-pruning strategy.

Measuring black-market transactions of prescription opioids: To test our third and fourth hypotheses and analyze the relationship between a focal state's PDMP-EHR integration and local black market prices for prescription opioids, we construct the variable $Price_per_MME_{ijk,t}$, representing the reported street price per MME unit for transaction i in state j and year t for main active ingredient k . By converting the original price per unit to an MME basis, we capture consumers' higher willingness to pay for more potent drugs.

We constructed this variable using a dataset of street drug transactions obtained from [StreetRx.com](https://www.streetrx.com), a crowdsourced research website launched by the Denver Health and Hospital Authority.⁷ [StreetRx.com](https://www.streetrx.com) collects user-submitted information on the street prices of diverted prescription and illicit drugs and allows anonymous viewing, posting, and rating of submissions. Each record includes details such as purchase date, location (state and optional city), product name, active ingredient, quantity (dosage form and strength), and price (per unit for prescription drugs or per transaction for illicit drugs). Our dataset comprises all reported street transactions since 2013, which ensures national representativeness by that year. To align with our ARCOS data, we end the sample in 2017. Consistent with our Hypothesis 1 analysis, we limit the sample to 66,813 transactions involving the same eight prescription opioids used in the construction of SOQ_{it} .

3.2 | Key Independent and Control Variables

Construction of Key Independent Variables: To test Hypotheses 1 and 2, we define treatment and control groups at the county level. A county is assigned to the control group if none of its home state's neighboring states has implemented PDMP-EHR integration. In this paper, unless otherwise specified, "a neighboring state of county i " refers to a state that shares a border with county i 's home state.

As shown in Figure 1, for a treated county i , the treatment starts when county i gets exposed to a neighboring state's PDMP-EHR integration. One challenge in this empirical setting is that the treated county i may have multiple neighboring states implementing PDMP-EHR integration over time. To address this issue and obtain a clearer interpretation of the causal effect, we apply a sample-pruning strategy. We identify the *earliest* neighboring state that implemented PDMP-EHR integration (so-called "first-neighboring" state) and denote by $T1_i$ the *start* year of county i 's first-neighboring state's PDMP-EHR integration. Similarly, we identify (where necessary) by $T2_i$ the year when the next neighboring state implements

PDMP-EHR integration. From year $T2_i$ onwards, the treated county is exposed to multiple neighboring states' PDMP-EHR integrations, so we denote by TN_i the year when the N th neighboring state implemented PDMP-EHR integration. We rely on this pruned sample (i.e., observations before $T2_i$) in our main analysis.

Our data show that about one-third of counties have multiple neighboring states implementing PDMP-EHR integration during the same "earliest" year $T1_i$. In that case, we further consider the role of travel distance and define county i 's first-neighboring state as the one whose border is geographically closest to county i 's centroid; the geographical distance between county i 's centroid and a neighboring state's border is measured by using geographic distance data from Holmes (1998). Thus, considering timing and distance, a treated county i would always have exactly one first-neighboring state, and different treated counties in the same state could have different first-neighboring states. Our treatment variable can capture the spillover effect from the first-neighboring state. This implicitly assumes that spillover effects from other neighboring states that adopt PDMP-EHR integration at year $T1_i$ are minimal. This assumption is supported by our empirical analysis in Section 4.1.4 and the results in Table 6.

To capture the above setup, we code a dummy variable, $Neighbor_integration_{it}$, to indicate whether county i 's first-neighboring state has implemented PDMP-EHR integration in year t . As an example, consider Anoka, a county in Minnesota. Minnesota has three adjacent states that adopted PDMP-EHR integration during the sample period: North Dakota in 2014, South Dakota in 2015, and Wisconsin in 2016. Hence, all counties in Minnesota, including Anoka, are considered treated starting in 2014. $Neighbor_integration_{it}$ for Anoka, as county i , equals one from 2014 and on and 0 for the years before 2014. Therefore, $Neighbor_integration_{it}$ is the key independent variable that allows us to estimate the spillover effect of the PDMP-EHR integration from county i 's first-neighboring state.

When multiple neighboring states qualify as first-neighboring states, we select the closest by distance, as travel distance significantly influences patients crossing state lines for medications. We identify PDMP-EHR integration dates following Wang (2021), with all dates listed in Table B1. For example, in New York, three neighboring states—Massachusetts, Vermont, and Pennsylvania—implemented PDMP-EHR integration in the same earliest year (2016). Using geographical proximity, we assigned Massachusetts as the first neighboring state for nine counties in New York, Pennsylvania for 40 counties, and Vermont for the remaining 13 counties.

To test Hypotheses 2 and 4, we define the variable $Sharing_{it}$ as a dummy variable indicating whether county i 's home state has been authorized to have access to the PDMP data of county i 's first-neighboring state in year t . We construct $Sharing_{it}$ this way to capture the spillover effect from the PDMP-EHR integration of county i 's first-neighboring state. PDMP data sharing status is determined by whether a state's PDMP program participates in the PMP InterConnect hub. The dates when each state began participating in PMP InterConnect were obtained from a document provided by the PMP InterConnect coordinator and reports from the PDMP Training and Technical Assistance Center (PDMP TTAC) of Brandeis University. Table B1 presents these dates. Table B2 shows no systematic pattern in the sequence of PDMP integration and data sharing across states.

Tables 1 and 2 report definitions and summary statistics of our variables.

3.3 | Other Control Variables

To account for PDMP-related changes likely affecting the supply of prescription opioids in focal county i , we control for several PDMP-related implementations or policy changes. First, we include $Focal_pdmp_{it}$ and $Neighbor_pdmp_{it}$, dummy variables indicating whether county i 's home or first neighboring states have implemented an operational PDMP in year t . We obtained state PDMP operating dates from the website of the PDMP TTAC⁸ (see Table B1), a reliable source used in prior studies (Meinhofer 2018; Patrick et al. 2016). Second, we control for the direct effect of focal states' PDMP-EHR integration through $Focal_integration_{it}$, a dummy variable indicating whether county i 's home state has implemented PDMP-EHR integration in year t . Third, we control for the presence of home and first-neighboring states' PDMP mandates through dummy variables $Focal_mandate_{it}$ and $Neighbor_mandate_{it}$. We obtained the PDMP mandatory use enactment dates from a prior study by Mallatt (2022)⁹ which we cross-validate with other studies (Buchmueller and Carey 2018; Sacks et al. 2021; Wang 2021) as well as manually verify for states with multiple mandate dates (see Table B1). Fourth, we control for the EHR adoption rate—an important enabler of PDMP-EHR integration—in hospitals in the focal and first-neighboring states through $Neighbor_ehr_{it}$ and $Focal_ehr_{it}$, representing the proportion of hospitals with basic EHRs in county i 's first-neighboring and focal states in year t . Data on EHR adoption are collected from the American Hospitals Association (AHA) Annual Survey IT Supplement, which defines as adopters of basic EHRs hospitals with computerized systems that have key recording capabilities (e.g., patient demographics, physician notes, medication lists etc.).

To account for confounding factors affecting county-level opioid sales, we control for several time-varying demographic and socioeconomic variables at the state and county levels. From the Bureau of Economic Analysis (BEA), we obtain data on population size, employment, average personal income at the state–county level, and the fractions of individuals with high-school and college education at the state level. Following the literature (Case and Deaton 2015), we include the proportions of the 15–24 and 25–34 age groups and the proportion of white Americans because age and ethnicity factors are associated with the risk of drug-related mortality. We obtain the data from the Surveillance

Epidemiology and End Results (SEER) database and aggregate individual-level age and ethnicity data at the county level.

4 | Research Design and Main Results

4.1 | Cross-Border Spillover Effect

4.1.1 | Empirical Identification

We employ the staggered difference-in-differences (DID) framework to estimate the spillover effect of county i 's first neighboring state's PDMP-EHR integration on county i . This identification strategy leverages the variation in timing across counties being affected by their first neighboring state adopting PDMP-EHR integration. Control counties that are not affected by any neighboring states' PDMP-EHR integration serve as counterfactuals for treated counties, representing potential outcomes in the absence of spillover effects. This methodology has been widely used in the literature (Buchmueller and Carey 2018; KC et al. 2022). We specify the main econometric models to test Hypotheses 1 and 2 as follows:

$$\begin{aligned} \ln(SOQ)_{it} = & \beta_1 Neighbor_integration_{it} \\ & + \beta_4 Neighbor_mandate_{it} \\ & + \beta_5 Neighbor_pdmp_{it} \\ & + \gamma X_{it} + \alpha_i + \tau_t + \epsilon_{it} \end{aligned} \quad (1)$$

$$\begin{aligned} \ln(SOQ)_{it} = & \beta_1 Neighbor_integration_{it} \\ & + \beta_2 Neighbor_integration_{it} \times Sharing_{it} \\ & + \beta_3 Sharing_{it} + \beta_4 Neighbor_mandate_{it} \\ & + \beta_5 Neighbor_pdmp_{it} \\ & + \gamma X_{it} + \alpha_i + \tau_t + \epsilon_{it} \end{aligned} \quad (2)$$

Equation (1) focuses on the main cross-border spillover effect without considering the moderating effect of $Sharing_{it}$, whereas Equation (2) is our full estimation specification, which includes $Sharing_{it}$ and its interaction term with $Neighbor_integration_{it}$. In Equation (2), β_1 captures the spillover effect of the first-neighboring state's PDMP-EHR integration on county i , the primary parameter of interest. The interaction term of $Neighbor_integration_{it}$ and $Sharing_{it}$ allows us to examine how the spillover effect would change depending on whether county i 's home state can access the first-neighboring state's PDMP data. Lastly, X_{it} represents the control variables defined in Section 3.3.

Because a neighboring state's overall PDMP implementation is likely determined by its controlled substance consumption level, the supplied opioid quantity of the first-neighboring state in the past year, $Neighbor_SOQ_{i(t-1)}$, could be correlated with $Neighbor_integration_{it}$. In addition, the first-neighboring state's supplied opioid quantity might be correlated with the number of its patients traveling across state borders to adjacent states to get prescriptions and thus correlated with county i 's consumption of prescribed opioids. Therefore, we also control $Neighbor_SOQ_{i(t-1)}$ in the estimation. We log-transform all other control variables except for proportion data and dummy variables. We control for county (α_i) and year (τ_t) fixed effects.

TABLE 1 | Variable name and definition.

Variable	Definition	Data source
SOQ	Supplied Opioid Quantity, measured by the total quantity (in MME equivalents) of the controlled opioid substances provided to the focal county in a given year.	ARCOS
Neighbor_SOQ	Yearly SOQ of the first-neighboring state	ARCOS
Neighbor_integration	Dummy variable indicating if a focal county's first-neighboring state integrated PDMP system with EHRs	Wang (2021)
Neighbor_mandate	Dummy variable indicating if a focal county's first-neighboring state mandates prescribers to check their state's PDMP before prescribing	Mallatt (2022)
Neighbor_pdmp	Dummy variable indicating if a focal county's first-neighboring state has an operational electronic PDMP	PDMP TTAC
Focal_integration	Dummy variable indicating if a focal state's PDMP system has been integrated with EHRs	Same as Neighbor_integration
Focal_mandate	Dummy variable indicating if a focal state mandates prescribers to check their state's PDMP before prescribing	Same as Neighbor_mandate
Focal_pdmp	Dummy variable indicating if a focal state has an operational electronic PDMP	Same as Neighbor_pdmp
Sharing	Dummy variable indicating whether a focal county's state and its first-neighboring state have agreed to share PDMP data through PMP Interconnect	NABP; PDMP TTAC
Neighbor_ehr	Portion of hospitals that have adopted basic EHRs in the neighboring state	AHA
Focal_ehr	Portion of hospitals that have adopted basic EHRs in the focal state	AHA
County_income	Annual average personal income of a county	BEA
County_population	Total population of a county	BEA
County_employment	Number of the employed population of a county	BEA
County_white_portion	Percentage of the white American population of a county	SSER
County_age15_24_portion	Percentage of the population aged from 15 to 24 in a county	SSER
County_age25_34_portion	Percentage of the population aged from 25 to 34 in a county	SSER
St_income	Annual average personal income of a state	BEA
St_population	Total population of a state	BEA
St_employment	Number of the employed population of a state	BEA
St_high_school_portion	Fraction of state population with high school level educational attainment	BEA
St_college_portion	Fraction of state population with college-level educational attainment	BEA
Price_per_MME	Unit price of reported drug in currency/per morphine-milligram-equivalence unit	StreetRx

4.1.2 | Parallel Trends Assumption and Sample Construction

We first assessed whether the parallel trends assumption, the primary assumption underlying our DID estimation strategy, holds (Angrist and Pischke 2008). In our context, this assumption implies the supplied opioid quantity in treated

counties follows a similar trend as in untreated counties prior to treatment.

To test this empirically, we employ a leads-and-lags (relative time) model to test the parallel trends assumption and examine post-treatment effects over several years. We specify this approach in Equation (3), using a model specification and

TABLE 2 | Summary statistics.

Variable	N	Mean	SD	Min	Median	Max
Panel A: Sample of cross-border spillover analysis						
Ln (SOQ)	22,556	9.653	1.651	3.186	9.574	15.111
Sharing	22,556	0.246	0.431	0	0	1
Neighbor_integration	22,556	0.316	0.465	0	0	1
Neighbor_mandate	22,556	0.234	0.423	0	0	1
Neighbor_pdmp	22,556	0.869	0.337	0	1	1
Focal_integration	22,556	0.112	0.315	0	0	1
Focal_mandate	22,556	0.182	0.386	0	0	1
Focal_pdmp	22,556	0.87	0.336	0	1	1
Neighbor_ehr	22,556	0.308	0.238	0	0.257	0.909
Focal_ehr	22,556	0.315	0.236	0	0.26	0.9
Ln (County_income)	22,556	10.502	0.244	9.387	10.477	12.206
Ln (County_population)	22,556	10.295	1.457	4.344	10.175	16.134
Ln (County_employment)	22,556	9.583	1.503	4.159	9.415	15.678
County_white_portion	22,556	0.868	0.159	0.09	0.938	1
County_age15_24_portion	22,556	0.13	0.032	0.045	0.123	0.406
County_age25_34_portion	22,556	0.116	0.021	0.048	0.114	0.283
Ln (St_income)	22,556	10.637	0.148	10.307	10.624	11.235
Ln (St_population)	22,556	15.576	0.923	13.235	15.609	17.493
Ln (St_employment)	22,556	15.038	0.893	12.861	15.08	16.976
St_high_school_portion	22,556	0.87	0.034	0.799	0.87	0.935
St_college_portion	22,556	0.275	0.045	0.173	0.271	0.567
Ln (Neighbor_SOQ)	22,556	14.722	0.942	12.488	14.931	16.431
Panel B: Sample of black-market spillover analysis						
Log(Price_per_MME + 1)	66,813	1.383	0.730	0.693	1.253	11.331

notation consistent with prior literature (Greenwood and Agarwal 2016).

$$\begin{aligned}
 \text{Ln(SPOQ)}_{it} = & \sum_{j=-5}^{-1} \beta_j \text{Neighbor_integration}(j)_{it} \\
 & + \sum_{j=1}^4 \beta_j \text{Neighbor_integration}(j)_{it} \quad (3) \\
 & + \delta_1 \text{Neighbor_mandate}_{it} \\
 & + \delta_2 \text{Neighbor_pdmp}_{it} \\
 & + \gamma X_{it} + \alpha_i + \tau_t + \varepsilon_{it}
 \end{aligned}$$

In this equation, $\text{Neighbor_integration}(j)_{it}$ equals one if year t is j years after the first implementation of PDMP-EHR integration among states neighboring county i , and zero otherwise (or for $j < 0$, it represents $-j$ years before the implementation). If none of the neighboring states implemented PDMP-EHR integration during the sample period, $\text{Neighbor_integration}(j)_{it}$ is set to zero.

The boundary period $j = -5$ represents five or more years before the integration. The reference period ($j = 0$), corresponding to the period of the first implementation of PDMP-EHR integration among the states neighboring county i , is omitted. The set of coefficients β_j allows us to validate the DID model by comparing pre-treatment trends between the treated and control groups. Equation (3) includes the same covariates $\text{Neighbor_mandate}_{it}$, $\text{Neighbor_pdmp}_{it}$ and X_{it} as in Equations (1) and (2). α_i and τ_t denote county and year fixed effects, respectively.

As shown in Column (1) of Table 3, the treatment effect persists for more than 4 years when using the full sample, which includes all counties and years. However, due to a violation of the parallel trends assumption in the pre-treatment period, to address the issue and make our analysis more rigorous, we removed any focal-county observations from our full sample as soon as a focal county got exposed to later neighboring states' PDMP-EHR integration to satisfy the parallel trends assumption, as described in Section 3.2. Column (2) of Table 3 reports the results from using the pruned sample. In this sample, the

TABLE 3 | Results from the relative time model.

Variables	(1)	(2)
	Full sample	Pruned sample
Neighbor_integration($t-5$)	−0.034** (0.013)	0.011 (0.016)
Neighbor_integration($t-4$)	−0.039*** (0.010)	0.001 (0.014)
Neighbor_integration($t-3$)	−0.048*** (0.008)	−0.016 (0.011)
Neighbor_integration($t-2$)	−0.034*** (0.005)	−0.011 (0.007)
Neighbor_integration($t-1$)	−0.022*** (0.003)	−0.005 (0.004)
Neighbor_integration(0)	Omitted base case	Omitted base case
Neighbor_integration($t+1$)	0.008*** (0.002)	0.011* (0.005)
Neighbor_integration($t+2$)	−0.003 (0.003)	−0.018* (0.009)
Neighbor_integration($t+3$)	0.012* (0.005)	−0.006 (0.012)
Neighbor_integration($t+4$)	0.019*** (0.004)	−0.051** (0.016)
Neighbor_mandate	0.043*** (0.005)	0.048*** (0.005)
Neighbor_pdmp	0.011+ (0.007)	0.039*** (0.005)
Controls	Yes	Yes
Observations	27,441	22,556
Within R^2	0.081	0.117
County number	3050	2922
County fixed effect	Yes	Yes
Year fixed effect	Yes	Yes

Note: For Tables 3–6, the row “Controls” indicates that we have controlled for a set of control variables as defined in Table 1. Specifically: Focal_integration, Focal_mandate, Focal_pdmp, Focal_ehr, Neighbor_ehr, County_income, County_population, County_white_portion, County_age15_24_portion, County_age25_34_portion, St_income, St_population, St_employment, St_high_school_portion, St_college_portion, Neighbor_SOQ_(t-1). + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

results show that all coefficients of β_j for the pre-treatment periods ($j < 0$) are statistically insignificant, suggesting that in the recent pre-treatment periods, there is essentially no systematic difference between the treated and untreated counties in terms of their supplied opioid quantity (Angrist and Pischke 2008; Bertrand et al. 2004). As pre-treatment trends are parallel in the pruned sample, we focus our interpretation on this sample's results in the rest of the paper.

4.1.3 | Main Results

The main results based on Equations (1) and (2) are presented in Table 4. Given that our analysis is at the county level, we cluster standard errors by county. First, in both Columns (1) and (2), we find that the estimated coefficients of *Focal_mandate* are negative and statistically significant. These results suggest that the PDMP mandate of a focal county's home state effectively reduces its supplied opioid quantity, consistent with recent studies (e.g., Buchmueller and Carey 2018; Sacks et al. 2021). More importantly, Columns (1) and (2) show that the estimated coefficients of *Neighbor_integration* are positive and statistically significant ($p < 0.001$). The results indicate that the PDMP-EHR integration of a focal county's first neighboring state is associated with an increase in the focal county's supplied opioid quantity, thus supporting Hypothesis 1.

Furthermore, the results in Column (2) show that the estimated coefficient of the interaction term *Neighbor_integration* × *Sharing* is negative and marginally significant ($p < 0.10$). This finding provides weak support for Hypothesis 2 and suggests that the spillover effect of a neighboring state's PDMP-EHR integration is likely moderated by whether the focal county's home state has access to the neighboring state's PDMP data.

The marginal significance of the interaction term between *Neighbor_integration* and *Sharing* could be driven by the conservative nature of our main analysis. The effect might be more pronounced in focal counties where the first neighboring state implementing PDMP-EHR integration is geographically closer and/or characterized by higher rates of EHR adoption. Accordingly, in the following section, we present additional analyses using EHR adoption rates and travel distance as moderators to further explore these relationships.

4.1.4 | Results on Heterogeneity in the Spillovers

This section sheds additional light on the underlying mechanisms that drive the observed cross-border spillover effect. Specifically, we focus on two possible factors: (i) EHR adoption rate and (ii) travel distance.

We first examine the differential spillover effects from neighboring states with high versus low EHR adoption rates. The premise is that a substantial EHR adoption rate in a neighboring state facilitates doctors' ability to check a patient's opioid purchases, thereby increasing the detection and prevention of over-consumption. Thus, the effectiveness of a neighboring state's

PDMP-EHR integration is likely to positively correlate with its EHR adoption rate. We expect stronger spillover effects when the neighboring state has higher EHR adoption.

To test this, we measure each state's EHR adoption rate by the proportion of hospitals with basic EHRs. We then divide our sample into two groups: counties with first-neighboring states having high EHR adoption rates and those with low EHR adoption rates. Using Equation (2), we estimate these subsamples separately and report the findings in Table 5. Results indicate that both the magnitude and significance of *Neighbor_integration* and its interaction term *Neighbor_integration* × *Sharing* are higher for counties whose neighboring states have high EHR adoption rates, bolstering our main results that PDMP integration in neighboring states leads to spillover effects.

The spillover effect of a neighboring state's PDMP-EHR integration likely depends on the geographical proximity of the focal county to the neighboring state's border, as travel distance significantly influences patients' choice of healthcare providers (Cepeda et al. 2013). We explore whether this spillover effect varies geographically, comparing counties closer to the border ("borderline" counties) with those further away ("interior" counties). We expect that the spillover effect is stronger in borderline counties due to easier travel access and is more effectively mitigated by access to the neighboring state's PDMP data.

To test this hypothesis, we focus on counties bordering neighboring states with high EHR adoption rates. We calculate the distance between each county's centroid and the neighboring state's border (Bloom et al. 2019; Holmes 1998). Borderline counties are defined as those in the bottom one-third quantile of distances from state borders, while interior counties are defined as those in the top one-third quantile. These quantiles were chosen to clearly distinguish counties that are most influenced by neighboring states (borderline) from those that are least

TABLE 4 | Spillover effect of neighboring state's PDMP-EHR integration.

Variables	(1)	(2)
Neighbor_integration	0.018*** (0.003)	0.020*** (0.003)
Neighbor_integration × Sharing		−0.008 ⁺ (0.005)
Sharing		−0.001 (0.004)
Neighbor_mandate	0.050*** (0.004)	0.050*** (0.005)
Neighbor_pdmp	0.042*** (0.005)	0.042*** (0.005)
Focal_integration	0.002 (0.005)	0.004 (0.005)
Focal_mandate	−0.057*** (0.005)	−0.056*** (0.004)
Focal_pdmp	0.009 ⁺ (0.005)	0.008 (0.005)
Controls	Yes	Yes
Observations	22,556	22,556
Within R^2	0.112	0.112
County number	2922	2922
County fixed effect	Yes	Yes
Year fixed effect	Yes	Yes

Note: Standard errors clustered at the county level are reported in parentheses.
⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

TABLE 5 | Heterogeneity in the spillover effect for EHR adoption rate in neighboring states.

Variables	(1)	(2)
	High EHR adoption rate in first-neighboring states	Low EHR adoption rate in first-neighboring states
Neighbor_integration	0.043*** (0.005)	0.018*** (0.004)
Neighbor_integration × Sharing	−0.024*** (0.006)	−0.010 (0.008)
Sharing	−0.012* (0.005)	0.032*** (0.007)
Neighbor_mandate	0.031*** (0.005)	0.062*** (0.010)
Neighbor_pdmp	0.027*** (0.008)	0.067*** (0.007)
Controls	Yes	Yes
Observations	12,903	9653
Within R^2	0.1	0.184
County number	1573	1349
County fixed effect	Yes	Yes
Year fixed effect	Yes	Yes

Note: ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

TABLE 6 | Heterogeneity in the spillover effect for borderline versus distant counties.

Variables	(1)	(2)
	Borderline counties	Interior counties
Neighbor_integration	0.062*** (0.011)	0.002 (0.012)
Neighbor_integration × Sharing	−0.039*** (0.012)	0.017+ (0.009)
Sharing	−0.030*** (0.009)	−0.016+ (0.009)
Neighbor_mandate	0.027** (0.009)	0.054*** (0.008)
Neighbor_pdmp	0.017 (0.013)	0.053*** (0.013)
Controls	Yes	Yes
Observations	4425	4006
R ²	0.13	0.146
County number	520	519
County fixed effect	Yes	Yes
Year fixed effect	Yes	Yes

Note: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

influenced (interior), allowing us to effectively capture potential spillover effects. We then create two subsamples: one including only borderline counties and the other only interior counties. We estimate these subsamples using Equation (2). The results are presented in Columns (1) and (2) of Table 6.

The findings reveal that the estimated coefficient for *Neighbor_integration* is positive and significant for borderline counties but not for interior counties, suggesting a stronger spillover effect in borderline regions. Moreover, the estimated coefficient of *Neighbor_integration* × *Sharing* is negative and significant for borderline counties, while it is only marginally significant for interior counties. These results indicate that both the spillover effect and the mitigating impact of interstate PDMP data sharing are more pronounced in borderline counties. This supports our hypothesis that the spillover effect is driven by cross-state border “doctor-shopping” behavior, where individuals seek prescriptions from multiple healthcare providers across state lines to avoid detection.

4.1.5 | Robustness Checks

We test the robustness of our results through various checks. First, we assess robustness concerning the construction of *Neighbor_Integration_{it}* since county *i* could be affected by spillovers from multiple neighboring states. Specifically, we construct a distance-weighted measure of *Neighbor_Integration_{it}*, to capture possible spillovers from all neighboring states of county *i* (Gaughan et al. 2015). Intuitively, we need to put more weight on a neighboring state *s* that is geographically closer to county *i*. We define *Distance_{si}* as the minimum geographical distance between county *i*'s centroid and the border of the neighboring state *s* and construct the key independent variable *Neighbor_Integration_{it}* as the following weighted measure: $Neighbor_Integration_{it} = \sum_{s \in S_i} w_s Neighbor_Integration_{st}$, where S_i is the set of county *i*'s neighboring states. *Neighbor_Integration_{st}* is a dummy variable and equals one if neighboring state *s* has

integrated PDMP into EHRs in year *t*, and $w_s = (1/Distance_{si}) / \sum_{s \in S_i} (1/Distance_{si})$ is the normalized weight for neighboring state *s*. We apply this weighted measure to all PDMP-related variables and socioeconomic controls to capture spillovers from all neighboring states implementing PDMP-EHR integration. Table B3 presents the estimated results using the weighted measures, which are consistent with our main results in Table 4. As expected, the magnitude of the estimated coefficient of *Neighbor_integration* in Table B3 is greater than that in Column (2) of Table 4, as *Neighbor_integration* here captures the spillover effect from all neighboring states.

Second, recall that if a focal county has several neighboring states which implemented PDMP-EHR integration, we have removed in the main analysis any observations after the second neighboring state's integration year to avoid overestimating the spillover effect of the first neighboring state's PDMP-EHR integration. We re-estimate here our models using the full sample of observations and present the results in Table B4. The results are largely consistent with our main results in Table 4.

Third, following prior literature (Chan and Ghose 2014), we conduct a placebo test to assess the possibility that our treatment variable picks up the significant spillover effect by chance. We run a regression using as a dependent variable the total number of claims of antibiotic drugs reimbursed by Medicare Part D while keeping the same model specification as our main model. We have collected the provider level number of claims of antibiotic drugs from the Center for Medicare & Medicaid Services and aggregated them into a county-level variable. The data starts in 2013 and covers until the end of our time window. Conceptually, PDMP-EHR integration in a focal county's neighboring state should not affect its own consumption of antibiotic prescriptions. We report the results in Table B5, showing that the estimated coefficient of *Neighbor_integration* is insignificant, and the magnitude is close to 0, implying that the spillover effect we identify is not likely to be detected by chance.

Lastly, we conduct an additional set of robustness tests to address concerns relating to the potential for false significance in our estimates or serial correlation in our dependent variable. We implement the test suggested by Bertrand et al. (2004). Specifically, we randomly assign the treatment to half of our sample. Then, we estimate the effects using the main model specification based on the pseudo-treatment assignment and store the estimated coefficients. We repeat the estimation procedures 500 times and compute the mean (μ of Random $\beta = -0.0001724$) and standard deviation (σ of Random $\beta = 0.028589$) of the pseudo-treatment coefficients. We compare our actual estimated coefficient of the spillover effect with the estimated coefficient of the pseudo treatment from the random assignments and find that the difference between the two is statistically significant ($z = 1.825$, $p < 0.1$). This suggests that it is highly unlikely that the spillover effect reported in our main analyses is detected by chance.

4.2 | Black-Market Spillover Effect

We test Hypotheses 3 and 4 using the following regression model:

$$\begin{aligned} \text{Price_per_MME}_{kmsst} = & \beta_1 \text{Focal_integration}_{st} \\ & + \beta_2 \text{Focal_integration}_{st} \times \text{Sharing}_{st} \\ & + \beta_3 \text{Focal_mandate}_{st} \\ & + \beta_4 \text{Focal_pdmp}_{st} \\ & + \gamma X_{st} + \alpha_s + \tau_t + \partial_m + \varepsilon_{kmsst} \end{aligned} \quad (4)$$

Equation (4) examines the relationship between a focal state's PDMP policies and the street prices of prescription opioids.

$\text{Price_per_MME}_{kmsst}$ represents the street price measure, defined as price of reported drug per morphine-milligram-equivalence unit for transaction k in state s in year t with the main active ingredient m . All the independent variables are defined consistently with the variables used in our analysis of the cross-border spillover effect. We control for the same state-level socioeconomic covariates, state fixed effect α_s , drug-specific fixed effects ∂_m , and year fixed effect τ_t . We do not control for county-level covariates, as approximately one-third of the transactions only report state-level location information.

Column (1) of Table 7 shows that the estimated coefficient on *Focal_integration* is positive but statistically insignificant when Equation (4) is applied to the full black-market transaction sample. The result also shows a statistically insignificant estimated coefficient on the interaction term between *Focal_integration* and *Sharing*. We further explore the possibility of black-market spillover effects using EHR adoption rates within focal states, as they are likely to enhance prescribers' ability to check a patient's prescription opioid usage. We argue that higher EHR adoption rates in focal states facilitate the surveillance role of PDMP-EHR integration and lead to increased local black-market transactions, driving higher street prices.

To test this, we conduct a subsample analysis similar to the one in Section 4.1.4, where we divide states into two groups: states with high EHR adoption rates and states with low EHR adoption rates. Using Equation (4), we estimate these subsamples separately and report the results in Table 7. Column (2) in Table 7 shows that PDMP-EHR integration in states with high EHR adoption rates is associated with a 1.6% increase in price per MME unit ($p < 0.5$) for black market opioids. Consistent with

TABLE 7 | Effect of PDMP-EHR integration on local black market transaction price.

Variables	DV: Log (price per MME + 1)		
	(1)	(2)	(3)
	Full sample	High EHR adoption rate states	Low EHR adoption rate states
Focal_integration	0.010 (0.010)	0.016* (0.006)	0.009 (0.015)
Focal_integration × Sharing	0.003 (0.014)	0.001 (0.010)	0.003 (0.021)
Sharing	−0.007 (0.011)	−0.002 (0.009)	−0.006 (0.019)
Focal_mandate	−0.011 (0.007)	−0.002 (0.010)	−0.011 (0.015)
Focal_pdmp	−0.039 (0.039)	−0.059 (0.040)	
Controls	Yes	Yes	Yes
Observations	66,813	37,051	29,762
R ²	0.636	0.639	0.632
Drug fixed effect	Yes	Yes	Yes
State fixed effect	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes

Note: Standard errors clustered at the state level are reported in parentheses. Significance levels are denoted as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, **** $p < 0.001$. The analysis controls for the same state-level socioeconomic variables used in the cross-border analysis. For this analysis, we assigned the EHR adoption rate during the focal integration year to each focal state as a split sample variable. To address missing values for states without PDMP integration until 2017, the EHR adoption rate from 2017 was used to impute these missing values. In Column (3), for this subsample, the Focal_pdmp variable remained constant. As a result, the coefficient on Focal_pdmp is omitted, as it is controlled for by the state fixed effects.

our argument, Column (3) in Table 7 shows that there is no significant spillover effect on street prices in states with low EHR adoption rates.

In summary, the findings provide nuanced support for Hypothesis 3, demonstrating that PDMP-EHR integration is positively associated with local street prices of prescription opioids when state-level EHR adoption rates are high. However, all the results show an insignificant estimated coefficient on the interaction term between a focal state's vertical PDMP-EHR integration and interstate data sharing. Therefore, we do not find evidence to support Hypothesis 4. Additionally, there is no evidence supporting a significant correlation between PDMP mandatory usage and black-market opioid prices.

5 | Discussion and Concluding Remarks

With this work we complement the literature at the interface of operations management and information systems by examining the nuanced operational value of HISs in curbing prescription opioid misuse. We investigate different integrations of the same information set (all leading to some form of increased informational transparency) and find that these can exhibit both desirable and undesirable effects. Vertical information integration that increases within-state informational transparency (i.e., PDMP-EHR integration) might unintentionally induce patients to cross state borders to obtain opioids. This cross-border flow can cause neighboring states to underestimate the effectiveness of their PDMP systems, partially explaining mixed findings in prior studies (Buchmueller and Carey 2018). Furthermore, the spillover effect of a neighboring state's PDMP-EHR integration intensifies when that state has a higher EHR adoption rate and is geographically closer to the focal county, suggesting that strategic cross-border “doctor-shopping” drives this behavior. These findings support the growing adoption of horizontal integration through interstate PDMP data sharing as a solution to address regional information asymmetries. Without data sharing via a network such as PMP InterConnect, prescribers are limited to monitoring patients' in-state activity, which may represent “only a small portion of the relevant patient activity” (Davis 2018).

Our estimates suggest a cross-border spillover effect equivalent to, on average, a 2.0% increase in yearly supplied quantity of prescription opioids. This estimate is likely to be a lower bound, as it is only based on the spillover effect from PDMP-EHR integration in a single state (i.e., the first-neighboring state). Table B3 suggests that the average weighted spillover effect across neighboring states is 5.1%. To put these numbers in perspective, we note that a PDMP mandate in the focal state typically reduces the annual supplied quantity of prescription opioids by 5.6%, whereas a mandate in the first-neighboring state increases it by 5%. These numbers align with prior research. KC et al. (2022) have estimated that “after states mandated PDMP use, opioid prescriptions decreased by 6.1% as intended” whereas Buchmueller and Carey (2018) indicate that PDMP mandates increase the rate of obtaining prescriptions from out-of-state prescribers by approximately a third of a standard deviation.

To emphasize that our estimates of the magnitude of cross-border spillover effects from PDMP-EHR integration are likely

to be lower bounds, we note also here that the implementation of PDMP-EHR integration is still lagging (Holmgren and Apathy 2020). Even though most states have legislated the integration of PDMP data into EHRs, which is reflected in our dataset, the process of rolling these features out is still ongoing and implementation of such functionality remains relatively low. According to the 2018 AHA Annual IT Survey, only about 30% of hospitals enable prescribers to check state PDMP data through their EHR. Moreover, the exact implementation characteristics vary across states and EHR systems. For example, in some implementations, the hospital EHR might automatically integrate data from the state PDMP without the need for an explicit query. Such differentiations and slow rollout pace result from the diversity of EHR systems, reluctance or limitations among vendors to provide necessary upgrades, associated costs, and, in some cases, inadequate prioritization (Baumgartner and Garretv 2020).

Our estimates also suggest that PDMP-EHR integration in states with high EHR adoption rates may induce a spillover to the black market for prescription opioids in the form of an approximately 1.6% average price increase. We do not find evidence of such a black-market spillover in states with low EHR adoption rates. Procuring from black markets is not the most common source for opioids. The Substance Abuse and Mental Health Services Administration (SAMHSA) national survey shows that only 7.9% of the 8.7 million people who misused pain relievers in 2021 obtained them from black markets (Substance Abuse and Mental Health Services Administration 2022). Ongoing research explores how PDMP implementation and mandates have the potential to spillover in the black market in the form of substitution toward illicit and potentially more lethal substances (Mallatt 2022). Exploring such potential effects of PDMP-EHR integration on drug diversion behavior is an interesting avenue for future research.

Our research has implications for designing prescription processes within HISs to support medical decisions and public health. Identifying cross-border doctor-shopping risks suggests the need for coordinated prescription practices beyond the state level. Well-designed HISs can enhance the integrity of opioid prescribing processes while reducing prescribers' operational burden. For example, PDMP-EHR integration with cross-state data sharing can improve prescription processes by automating alerts for overlapping prescriptions, recommending nonopioid alternatives, flagging risk factors (e.g., history of substance use or mental health conditions), and applying soft limits on opioid prescriptions based on patient risk profiles. Achieving these capabilities requires safe, interoperable, and privacy-sensitive systems, supported by adequate resources (Devaraj et al. 2013; Wang et al. 2024). PDMP-EHR integration might also facilitate opioid supply chain tracking, helping reduce excess supply and black market diversion risks (Maruchek et al. 2011). Our research encourages policy discussions on the spillover effects of regional regulations on controlled goods, suggesting that policy disparities across regions may induce spatial spillovers, potentially reducing policy effectiveness and creating unintended health, economic, and social impacts.

Our study, like similar works, has limitations that suggest directions for future research. First, our measures focus on the first

neighboring state to implement PDMP-EHR integration, potentially underestimating spillover effects from multiple neighboring states. Future studies could use a spatial econometric model to capture spillover effects from multiple neighboring states more precisely. Second, key variables like PDMP-EHR integration are at the state level but may vary by county; emerging hospital-level data could refine these measures. Additionally, our county-level opioid consumption data could be complemented by patient- or doctor-specific data for deeper insights into doctor-shopping behavior. Fourth, a more granular black-market analysis is a promising direction of future research and could be possible if individual-level transaction data becomes available, as aggregated data likely dilute the effect of policy interventions on illicit transactions. Finally, while hospitals are major sources of ambulatory care and opioid prescribing (Holmgren and Apathy 2020), further research could examine the impact of PDMP-EHR integration in nonhospital prescribing settings as integration becomes more widespread.

Acknowledgments

We sincerely thank the three anonymous reviewers, associate editor, and special issue editors for their detailed and constructive comments on earlier versions of the paper. We also thank Elliot Bendoly, co-editor-in-chief, for his support and guidance throughout the review process. Yufei Shen acknowledges funding support from FCT (UIDB/00124/2020, UIDP/00124/2020, UID/00124, POR Lisboa (PINFRA/22209/2016), POR Norte (PINFRA/22209/2016). Dorothy Jiang acknowledges funding support from the Behavioral Research Assistance Grant, the C.T. Bauer College of Business, University of Houston. Dimitrios Andritsos acknowledges funding support from the HEC Paris Foundation, Qatar Foundation, and Labex Ecodec: Investissements d'Avenir (ANR-11-IDEX-0003/Labex Ecodec/ANR-11-LABX-0047). Xitong Li acknowledges funding support from the HEC Foundation at HEC Paris and research fellowship from Hi! PARIS, a joint research center between HEC Paris and Institut Polytechnique de Paris.

Endnotes

¹ See the example of CharmHealth EHR user guide and watch the video about how prescribers can easily access PDMP data within an EHR: <https://www.charmhealth.com/resources/addons/pdmp.html>.

² A focal county can be located either along the border or at the interior of its own state. In Section 4.1.4, we explore whether the impact of a neighboring state's PDMP-EHR integration differs based on the focal county's location.

³ See the price comparison of popular opioids at: <https://www.addictioncenter.com/drugs/how-much-do-drugs-cost/>.

⁴ See the official website of the National Motor Vehicle Title Information System: <https://vehiclehistory.bja.ojp.gov/>.

⁵ See the official website of the National Instant Criminal Background Check System: <https://www.fbi.gov/how-we-can-help-you/more-fbi-services-and-information/nics/about-nics>.

⁶ The detailed information of the history of PMP InterConnect and its current usage can be found at their website: <https://pdmpworks.org/faq/#pdmp>.

⁷ We are grateful to StreetRx.com for supporting our research and providing the dataset since 2013.

⁸ The state PDMP profiles are available at <https://www.pdmpassist.org/State>.

⁹ We are grateful to Dr. Mallatt who kindly provided the list of dates when each state mandated that prescribers must check their state's

PDMP under certain circumstances and the corresponding information sources.

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Appendix A

TABLE A1 | Prior empirical research on PDMP and unanswered questions.

Research focus	Article	Main findings
Direct (i.e., within-state) Effects of PDMP Implementation, Mandates, and/or Interstate Data Sharing	Paulozzi et al. (2011)	PDMP implementation is not significantly with lower rate of opioid overdose mortality or lower rate of opioids consumption.
	Reifler et al. (2012)	PDMP implementation mitigates opioid misuse and abuse. But, effect is small.
	Brady et al. (2014)	PDMP implementation had no significant impact on per-capita opioids dispensed.
	Rutkow, Chang, et al. (2015)	PDMP implementation in Florida was associated with modest decreases in opioid prescribing concentrated among providers with higher baseline opioid volume.
	Bao et al. (2016)	PDMP implementation is associated with a sizable reduction in the prescribing of Schedule II opioids.
	Patrick et al. (2016)	PDMP implementation was associated with a reduction of opioid-related death rate. The effect is stronger if PDMP monitors more numbers of drugs and updates data more frequently.
	Chang et al. (2016)	PDMP implementation in Florida was associated with statistically significant relative reductions in several prescribing outcomes among high-risk prescribers, with virtually no effect on lower risk prescribers.
	Dowell et al. (2016)	PDMP mandates, together with pain clinic laws, were significantly associated with both decreased amounts of opioids prescribed and with decreased prescription opioid overdose deaths.
	Meara et al. (2016)	PDMP implementation had no significant effect on prescription opioid use among disabled Medicare beneficiaries.
	Wen et al. (2017)	PDMP mandates were associated with population-adjusted numbers of Schedule II opioid prescriptions received by Medicaid enrollees and amounts of Medicaid spending on these prescriptions.
	Bao et al. (2018)	PDMP mandates and delegation laws may reduce 6% to 9% high-risk opioid prescriptions. Interstate PDMP data sharing is not associated with meaningful changes in overall high-risks opioid prescriptions.
	Buchmueller and Carey (2018)	PDMP implementation alone had no significant effects, but PDMP mandates were associated with a reduction in measures of misuse in Medicare Part D.
	Haffajee et al. (2018)	PDMP mandates proportion of commercially insured adults who receive opioid prescriptions as well as the strength of those prescriptions.
	Meinhofer (2018)	PDMP implementation alone had no significant effects, but PDMP mandates were associated with a significant reduction in prescription opioid quantities and related overdose deaths.
	Pauly et al. (2018)	PDMP implementation reduces risk of prescription opioid-related poisoning. The effects are stronger when PDMPs monitor more schedules and require frequent data reporting.
	Greco et al. (2019)	PDMP implementation alone had no significant effects, but PDMP mandates significantly reduced treatment admissions for both prescription drug abuse and illicit opioid abuse.
	Wen et al. (2019)	PDMP mandates were associated with a reduction in the rates of opioid prescriptions and opioid-related inpatient stays and ED visits.
	Wen et al. (2020)	PDMP use mandates were associated with meaningful reductions in opioid-related hospital events among privately insured and Medicare Advantage adults with prescription opioid use.
	Kim (2021)	PDMP mandates increased heroin death rates.
	Sacks et al. (2021)	PDMP mandates reduced opioids prescribed to new patients.
	KC et al. (2022)	PDMP mandates reduced opioid prescriptions but increased more lethal illicit substitutes such as heroin.
	Gihleb et al. (2022)	PDMP implementations alone had significant effects on foster care admissions. However, PDMP mandates reduced child removals by 10%.
	Mallatt (2022)	PDMP implementations caused increases in heroin possession and dealers.
	Alpert et al. (2024)	PDMP mandates significantly reduced opioid prescription rates. The reduction in prescriptions is largely driven by patients presenting with conditions inappropriate for opioids.

(Continues)

TABLE A1 | (Continued)

Research focus	Article	Main findings
Cross-border Spillover Effects of PDMP implementation, or PDMP Mandate	Buchmueller and Carey (2018)	PDMP mandates have marginally significant effects in increasing out-of-state opioid consumption among Medicare Part D beneficiaries.
	Guth and Zhang (2022)	PDMP implementation reduces a state's opioid sales and overdose deaths but increases opioid sales and overdose deaths in neighboring counties on the other side of the state border.
Direct Effect of PDMP-EHR Integration	Neprash et al. (2022)	This cluster randomized clinical trial study found that PDMP integration led to a 60% increase in clinicians' PDMP queries.
	Hartung et al. (2024)	PDMP integration leads to a 2.74-fold increase in PDMP queries of prescribers. This effect is stronger among prescribers with lower PDMP use before the integration.
	Wang (2021)	PDMP integration reduces opioid-related inpatient morbidity, but no effect on mortality. Interstate data sharing alone has no effect, but can strengthen PDMP integration's effect.
Unanswered questions		
1. What are the potential cross-border and black-market spillover effects resulting from vertical Prescription Drug Monitoring Program (PDMP) data integration? 2. How might horizontal cross-state PDMP data sharing influence these spillover effects? 3. How do boundary conditions—such as the proximity of a focal county to a state border and the Electronic Health Record (EHR) adoption rate in a neighboring state implementing vertical PDMP data integration—moderate these spillover effects?		

Appendix B**TABLE B1** | Dates of state PDMP laws and complementary practices.

State	PDMP	Integration	Mandate	Interstate data sharing
Alabama	pre-2007			2017q1
Arizona	2008q3	2017q2	2017q4	2012q2
Arkansas	2013q1	2015q3		2013q4
California	pre-2007		2017q1	
Colorado	2007q3	2015q3		2013q2
Connecticut	2008q3		2015q4	2012q1
Delaware	2012q1		2012q3	2013q4
Florida	2011q3			
Georgia	2013q3			2016q3
Idaho	pre-2007	2014q2		2014q1
Illinois	pre-2007			2013q1
Indiana	pre-2007	2017q3	2014q3	2011q3
Iowa	2009q1			2015q2
Kansas	2011q1	2013q4		2012q2
Kentucky	pre-2007		2012q3	2013q1
Louisiana	2008q4	2015q4	2014q3	2013q1
Maine	pre-2007	2014q2	2017q1	2017q2
Maryland	2013q3	2013q4		2015q3
Massachusetts	pre-2007	2016q1	2014q3	2016q3
Michigan	pre-2007	2017q2		2012q1
Minnesota	2010q1			2013q4

(Continues)

TABLE B1 | (Continued)

State	PDMP	Integration	Mandate	Interstate data sharing
Mississippi	pre-2007	2015q2		2013q4
Missouri				
Montana	2012q1			2017q1
Nebraska	2011q2	2011q2		
Nevada	pre-2007	2015q1	2007q3	2014q1
New Hampshire	2014q3		2016q1	2016q4
New Jersey	2011q3		2015q4	2014q2
New Mexico	pre-2007	2015q2	2012q3	2012q3
New York	pre-2007	2016q1	2013q3	2016q2
North Carolina	2007q3			
North Dakota	2007q3	2014q2	2014q4	2012q1
Ohio	pre-2007	2015q4	2011q4	2011q3
Oklahoma	pre-2007	2014q2	2015q4	2015q1
Oregon	2011q2	2017q3		
Pennsylvania	pre-2007	2016q3	2016q4	2017q2
Rhode Island	pre-2007		2016q2	2014q4
South Carolina	2008q1	2015q3	2016q2	2012q1
South Dakota	2011q4	2015q3		2013q1
Tennessee	pre-2007	2015q4	2013q2	2013q3
Texas	pre-2007	2016q1	2015q3	2016q3
Utah	pre-2007		2016q2	2014q3
Vermont	2009q1	2016q1	2013q2	2016q2
Virginia	pre-2007	2015q2	2015q3	2011q3
Washington	2011q4	2014q4		
West Virginia	pre-2007	2015q3	2012q2	2014q1
Wisconsin	2013q2	2016q1	2017q2	2013q4
Wyoming	pre-2007			

Note: This table reports the appropriate PDMP-related dates (year and quarter) for our analytical sample, starting from 2008 and ending in 2017. Effective dates before 2007 are coded as pre-2007. A blank indicates a focal state still needs to implement PDMP/Integration/Mandate/Interstate data sharing by the end of 2017.

Source: PDMP implementation dates are collected from the website of PDMP TTAC and cross validated with Wang (2021). PDMP integration dates are collected from Wang (2021). Dates of PDMP-mandated usage law enactments are provided by Mallatt (2022), and compared the mandate dates with other papers (Buchmueller and Carey 2018; Wang 2021). Dates of each state joining PMP InterConnect are provided by the staff working at the NABP and serving as the liaison of PMP InterConnect.

TABLE B2 | Summarized pattern of PDMP-related enhancements.

The pattern of PDMP enhancements	Number of states
No PDMP	1
Only PDMP	3
Only integration	4
Only mandate	1
Only data sharing	6
Sharing, integration, mandate	11
Sharing, mandate, integration	7
Mandate, sharing, integration	7
Mandate, integration, sharing	4
Integration, mandate, sharing	2
Integration, sharing, mandate	2
Total	48

Note: Based on the list of dates in Table B1, this table reports the number of states that adopted PDMP-related enhancements following a specific sequence during our sample period from 2008 to 2017. For example, we identified 11 state PDMPs adopting interstate data sharing first, then integration, and mandated usage later.

TABLE B3 | Spillover effect of all neighboring states' PDMP integration using weighted measure.

Variables	(1) Weighted by distance
Neighbor_integration	0.051*** (0.009)
Neighbor_integration × Sharing	−0.016 ⁺ (0.009)
Sharing	0.005 (0.005)
Neighbor_mandate	0.021** (0.008)
Neighbor_pdmp	−0.001 (0.012)
Controls	Yes
Observations	27,441
Within R ²	0.062
County number	3050
County fixed effect	Yes
Year fixed effect	Yes

Note: ⁺ $p < 0.10$, $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

TABLE B4 | Spillover effect of neighboring state's PDMP integration using full sample.

Variables	(1)	(2)
Neighbor_integration	0.031*** (0.004)	0.043*** (0.005)
Neighbor_integration × Sharing		−0.031*** (0.006)
Sharing		0.015*** (0.004)
Neighbor_mandate	0.043*** (0.004)	0.043*** (0.005)
Neighbor_pdmp	0.011 (0.008)	0.008 (0.008)
Controls	Yes	Yes
Observations	27,441	27,441
Within R ²	0.078	0.08
County number	3050	3050
County fixed effect	Yes	Yes
Year fixed effect	Yes	Yes

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

TABLE B5 | Placebo test using the number of antibiotic claims.

Variables	(1)
Neighbor_integration	−0.002 (0.001)
Neighbor_integration × Sharing	0.002 (0.002)
Sharing	0.002 (0.002)
Neighbor_mandate	−0.005** (0.002)
Controls	Yes
Observations	10,872
Within R ²	0.782
County number	2922
County fixed effect	Yes
Year fixed effect	Yes

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The variable Neighbor_pdmp has been removed because of collinearity.